From Pi to PC:
My Self-Hosting Journey

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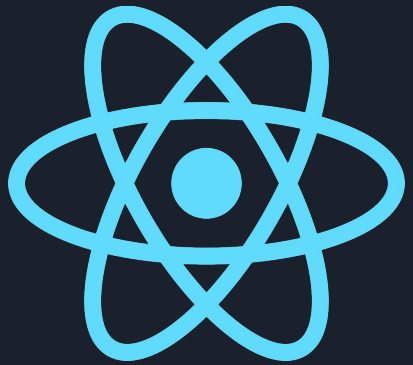
Who even is this guy?

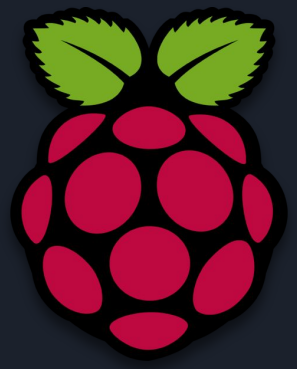
Ollscoil na Gaillimhe
University of Galway

arm

Primary Motivator: University Assignment

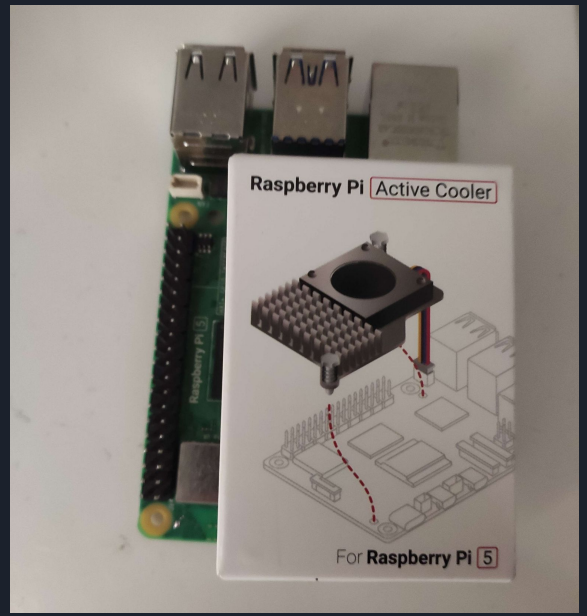
- Software Engineering 2 Group Project
- No Hosting Strictly Required
- Good Opportunity

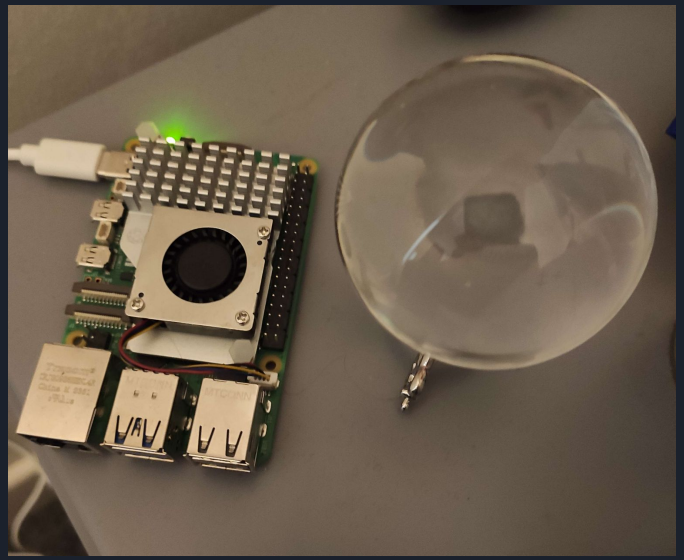




Installation and Setup

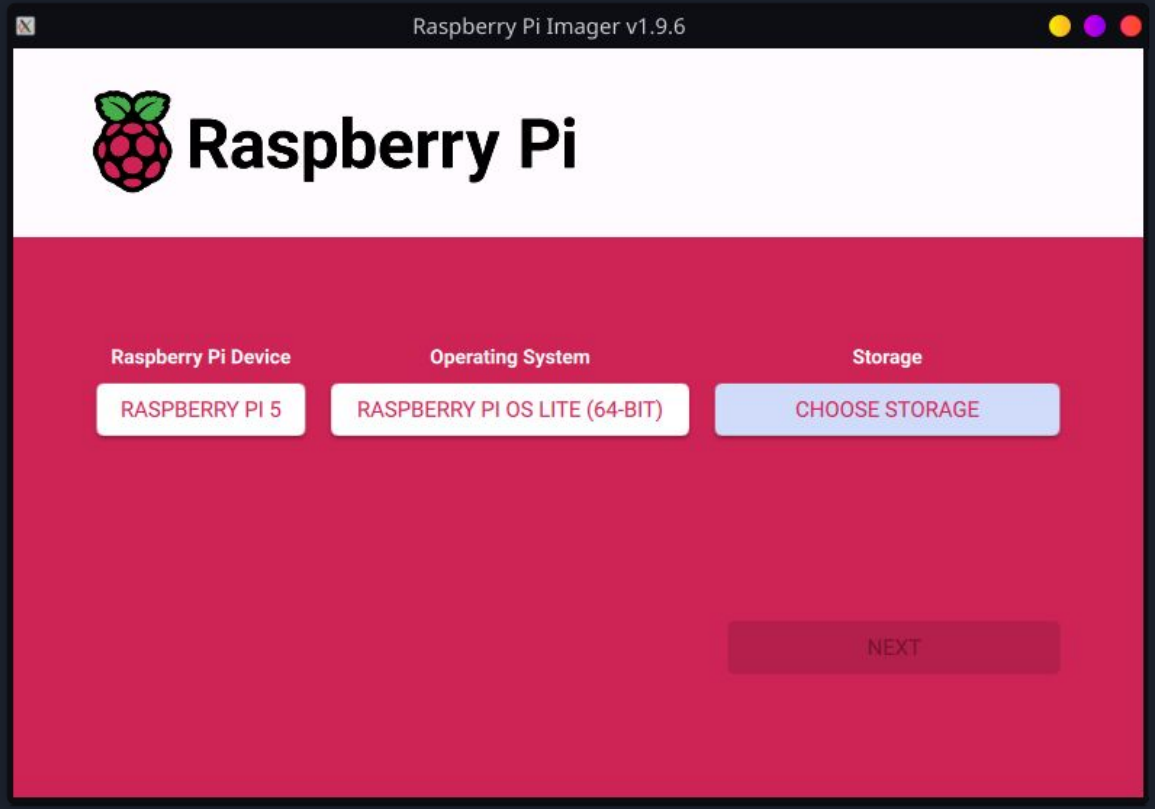
- Raspberry Pi 5 8 GB RAM variant, official Active Cooler
- Installed Headless PiOS on MicroSD (128 GB storage)





He lives next to The Orb

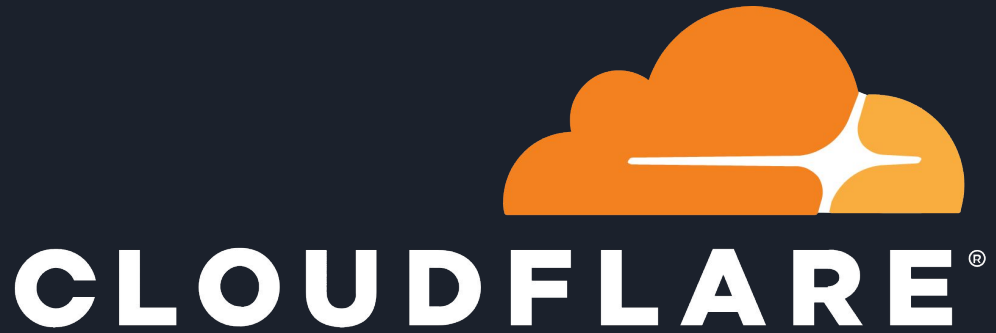
Installation and Setup



https://www.raspberrypi.com/documentation/computers/getting-started.html#raspberry-pi-imager

Public Internet Access

- Cloudflare Domain, Cloudflare Tunnel, Cloudflare Dashboard (no public IP)
- Online: Zero Trust Dashboard -> Connectors -> Set up a Tunnel
- On Device: Run ‘cloudflared’ with tunnel token found online



https://developers.cloudflare.com/cloudflare-one/networks/connectors/cloudflare-tunnel/get-started/create-remote-tunnel/

Public Internet Access

https://domains.cloudflare.com/

look-at-me-mom-im-selfhosting.org | Search

New .ai domains now available

look-at-me-mom-im-selfhosting.org — $7.50 Renews at $10.13 — Purchase

Suggested domain names

The following is a list of suggestions that may also be available.

Sort by Relevance

look-at-me-mom-im-selfhosting.com	$10.46 Renews at $10.46	Confirm
look-at-me-mom-im-selfhosting.net	$11.86 Renews at $11.86	Confirm
lookatmemomimselfhosting.org	$7.50 Renews at $10.13	Confirm

Services

- SWE project - Spring Boot backend, database, React.js frontend + javadoc (nginx), Jenkins, Authentik
- RAM fills up every few days and the Pi crashes :(

authentik

Docker Setup

- Docker Compose, multiple services sharing single cloudflared, nested DinD for Backend+db in a separate internal compose project
- Cloudflared - prebuilt image (*cloudflare/cloudflared*), no install needed
- Refactoring to get rid of DinD got rid of prod database :(

Refactoring Notes:

- Docker Compose multi-step build to optimise image size
- *include:* directive in Compose yaml -> multiple reusable config files
- Database container on known volume on disk (+backed up)

```
cloudflared:
  image: cloudflare/cloudflared:latest
  restart: unless-stopped
  command: tunnel run
  mem_limit: "128m"
  env_file:
    - ".env"
  environment:
    - "TUNNEL_TOKEN=${CLOUDFLARE_TUNNEL_TOKEN}"
  volumes:
    - /etc/localtime:/etc/localtime:ro
```

Current State

- Migrated to old Dell Optiplex desktop over Christmas (lives in parents' house abroad)
- Hosting stuff for family and friends (for example Filebrowser!)
- Working on VPS setup using KVM/QEMU and libvirt

Lessons Learned

1. Keep GUI Off Servers!
   Keep SSH, Screen and Keyboard Handy
2. Proper Cooling and Ventilation is Good to Have
3. Cloudflare Tunnels Need a Domain,
   Everything Else Easy
   (a *.org* costs less than a takeaway meal nowadays)
4. Strict Resource Limits for Docker Containers!
5. Docker Compose is Your Friend,
   Multi-step Builds Run Faster
   (don't forget to back up any production databases)

GitHub Gist with Information and Resources

Thank You!

Questions?